

SECTION 1: LINE SHAFT DIAMETER — TORSION ONLY

System Parameters: Speed $N = 200$ r.p.m., Power $P = 20$ kW = 20×10^3 W, Allowable Shear Stress $\tau = 42$ MPa = 42N/mm².

Design Protocol: Compute transmitted torque T from power-speed relation, apply torsion formula $T = \frac{\pi}{16}d^3\tau$ to size solid shaft diameter d .

1. Transmitted Torque (T)

Power-to-torque conversion at given speed

$$\begin{aligned} T &= \frac{P \times 60}{2\pi N} \\ &= \frac{20 \times 10^3 \times 60}{2\pi \times 200} \\ &= 955 \text{ N}\cdot\text{m} = \mathbf{955 \times 10^3 \text{ N}\cdot\text{mm}} \end{aligned}$$

2. Shaft Diameter (d)

Torsion strength formula for solid circular shaft

$$\begin{aligned} 955 \times 10^3 &= \frac{\pi}{16} \times d^3 \times \tau = \frac{\pi}{16} \times d^3 \times 42 = 8.25d^3 \\ d^3 &= \frac{955 \times 10^3}{8.25} = 115733 \\ d &= 48.7 \text{ mm} \\ &\Rightarrow \mathbf{\text{Adopt } d = 50 \text{ mm}} \end{aligned}$$

Design Output Summary

Line Shaft: P = 20 kW, N = 200 rpm, tau = 42 MPa | T = 955 kN-mm | d = 50 mm

SECTION 2: SOLID SHAFT — PEAK TORQUE FACTOR

System Parameters: Power $P = 1 \text{ MW} = 1 \times 10^6 \text{ W}$, Speed $N = 240 \text{ r.p.m.}$, Peak Torque $T_{\text{max}} = 1.2T_{\text{mean}}$, Allowable Shear Stress $\tau = 60 \text{ MPa} = 60\text{N/mm}^2$.

Design Protocol: Compute mean torque from power equation, apply peak factor 1.2 to find maximum torque, size diameter using torsion formula.

1. Mean Torque (T_{mean}) Power-to-torque conversion at rated speed	$T_{\text{mean}} = \frac{P \times 60}{2\pi N}$$= \frac{1 \times 10^6 \times 60}{2\pi \times 240}$$= 39784 \times 10^3 \text{ N}\cdot\text{mm}$
2. Maximum Torque (T_{max}) Peak torque with 20% overload factor	$T_{\text{max}} = 1.2T_{\text{mean}}$$= 1.2 \times 39784 \times 10^3$$= 47741 \times 10^3 \text{ N}\cdot\text{mm}$
3. Shaft Diameter (d) Torsion formula using maximum torque	$47741 \times 10^3 = \frac{\pi}{16} \times d^3 \times 60 = 11.78d^3$$d^3 = \frac{47741 \times 10^3}{11.78} = 4053 \times 10^3$$d = 159.4 \text{ mm}$$\Rightarrow \text{Adopt } d = 160 \text{ mm}$
Design Output Summary	Solid Shaft: P = 1 MW, N = 240 rpm, Tmax = 1.2 Tmean Tmax = 47741 kN-mm d = 160 mm

SECTION 3: SOLID AND HOLLOW SHAFT WITH FACTOR OF SAFETY

System Parameters: Power $P = 20 \text{ kW} = 20 \times 10^3 \text{ W}$, Speed $N = 200 \text{ r.p.m.}$, Ultimate Shear Stress $\tau_u = 360 \text{ MPa}$, Factor of Safety F.S. = 8, Hollow Ratio $k = \frac{d_i}{d_o} = 0.5$.

Design Protocol: Compute allowable shear stress from F.S., find torque, size solid shaft diameter d , then size hollow shaft using $(1 - k^4)$ reduction factor.

1. Allowable Shear Stress (τ) Dividing ultimate shear stress by factor of safety	$\tau = \frac{\tau_u}{\text{F.S.}} = \frac{360}{8} = 45 \text{N/mm}^2$
2. Transmitted Torque (T) Power-speed torque relation	$T = \frac{P \times 60}{2\pi N} = \frac{20 \times 10^3 \times 60}{2\pi \times 200} = 955 \times 10^3 \text{ N}\cdot\text{mm}$
3. Solid Shaft Diameter (d) Torsion formula for solid circular cross-section	$955 \times 10^3 = \frac{\pi}{16} \times d^3 \times 45 = 8.84d^3$$d^3 = \frac{955 \times 10^3}{8.84} = 108032$$d = 47.6 \text{ mm}$$\Rightarrow \text{Adopt } d = 50 \text{ mm}$
4. Hollow Shaft Outside Diameter (d_o) Torsion formula with hollow factor $(1 - k^4)$	$955 \times 10^3 = \frac{\pi}{16} \times 45 \times (d_o)^3 [1 - (0.5)^4] = 8.3(d_o)^3$$(d_o)^3 = \frac{955 \times 10^3}{8.3} = 115060$$d_o = 48.6 \text{ mm}$$\Rightarrow \text{Adopt } d_o = 50 \text{ mm}$
5. Hollow Shaft Inside Diameter (d_i) Applying diameter ratio $k = 0.5$	$d_i = k \cdot d_o = 0.5 \times 50 = 25 \text{ mm}$

Design Output Summary

Solid d = 50 mm | Hollow do = 50 mm, di = 25 mm | tau_allow = 45 MPa | T = 955 kN-mm

SECTION 4: RAILWAY AXLE UNDER PURE BENDING

System Parameters: Axle Load $W = 50 \text{ kN} = 50 \times 10^3 \text{ N}$ (per axle box), Overhang $L = 100 \text{ mm}$, Rail Gauge $x = 1.4 \text{ m}$, Allowable Bending Stress $\sigma_b = 100 \text{ MPa} = 100\text{N/mm}^2$.

Design Protocol: Compute maximum bending moment at the wheels as $M = W \cdot L$, apply bending formula $M = \frac{\pi}{32}d^3\sigma_b$ to find axle diameter.

1. Maximum Bending Moment (M) Bending moment at wheel location due to overhang load	$M = W \cdot L = 50 \times 10^3 \times 100 = \mathbf{5 \times 10^6 \text{ N}\cdot\text{mm}}$
2. Axle Diameter (d) Bending strength formula for solid circular section	$5 \times 10^6 = \frac{\pi}{32} \times d^3 \times \sigma_b = \frac{\pi}{32} \times d^3 \times 100 = 9.82d^3$ $d^3 = \frac{5 \times 10^6}{9.82} = 0.51 \times 10^6$ $d = 79.8 \text{ mm}$ $\Rightarrow \text{Adopt } d = 80 \text{ mm}$
Design Output Summary	Railway Axle: W = 50 kN, L = 100 mm overhang M = 5 MN-mm sigma_b = 100 MPa d = 80 mm

SECTION 5: SHAFT UNDER COMBINED BENDING AND TORSION

System Parameters: Bending Moment $M = 3000 \text{ N}\cdot\text{m} = 3 \times 10^6 \text{ N}\cdot\text{mm}$, Torque $T = 10000 \text{ N}\cdot\text{m} = 10 \times 10^6 \text{ N}\cdot\text{mm}$, Ultimate Tensile Stress $\sigma_{t \text{ u}} = 700 \text{ MPa}$, Ultimate Shear Stress $\tau_u = 500 \text{ MPa}$, Factor of Safety F.S. = 6.

Design Protocol: Compute allowable stresses; apply (1) max shear stress theory via equivalent twisting moment $T_e = \sqrt{M^2 + T^2}$; apply (2) max normal stress theory via equivalent bending moment $M_e = \frac{M+T_e}{2}$; take larger resulting diameter.

1. Allowable Stresses (σ_b, τ) Material strengths reduced by factor of safety	$\sigma_b = \frac{\sigma_{t \text{ u}}}{\text{F.S.}} = \frac{700}{6} = 116.7\text{N/mm}^2$$\tau = \frac{\tau_u}{\text{F.S.}} = \frac{500}{6} = 83.3\text{N/mm}^2$
2. Equivalent Twisting Moment (T_e) Max shear stress theory: resultant of bending and torsion	$T_e = \sqrt{M^2 + T^2} = \sqrt{(3 \times 10^6)^2 + (10 \times 10^6)^2} = \mathbf{10.44 \times 10^6 \text{ N}\cdot\text{mm}}$
3. Diameter by Max Shear Stress Theory (d_1) Torsion formula applied to T_e	$10.44 \times 10^6 = \frac{\pi}{16} \times d^3 \times 83.3 = 16.36d^3$$d^3 = \frac{10.44 \times 10^6}{16.36} = 0.636 \times 10^6$$d = \mathbf{86 \text{ mm}}$
4. Equivalent Bending Moment (M_e) Max normal stress theory: combined resultant	$M_e = \frac{1}{2}[M + T_e] = \frac{1}{2}(3 \times 10^6 + 10.44 \times 10^6) = \mathbf{6.72 \times 10^6 \text{ N}\cdot\text{mm}}$
5. Diameter by Max Normal Stress Theory (d_2) Bending formula applied to M_e	$6.72 \times 10^6 = \frac{\pi}{32} \times d^3 \times 116.7 = 11.46d^3$$d^3 = \frac{6.72 \times 10^6}{11.46} = 0.586 \times 10^6$$d = \mathbf{83.7 \text{ mm}}$

6. Governing Diameter Larger value from two failure theories governs	$d = \max(86, 83.7) = 86 \text{ mm}$ $\Rightarrow$ Adopt $d = 90 \text{ mm}$
Design Output Summary	$M = 3 \text{ MN-mm}, T = 10 \text{ MN-mm} \mid T_e = 10.44 \text{ MN-mm} \Rightarrow d_1 = 86 \text{ mm} \mid M_e = 6.72 \text{ MN-mm} \Rightarrow d_2 = 83.7 \text{ mm} \mid$ Governing $d = 90 \text{ mm}$

SECTION 6: SPUR GEAR SHAFT — COMBINED BENDING AND TORSION

System Parameters: Power $P = 7.5\text{ kW} = 7500\text{ W}$, Speed $N = 300\text{ r.p.m.}$, Gear PCD $D = 150\text{ mm}$, Bearing Span $L = 200\text{ mm}$, Allowable Shear Stress $\tau = 45\text{ MPa}$, Pressure Angle $\alpha = 20^\circ$.

Design Protocol: Find torque T ; compute tangential gear force $F_t = 2\frac{T}{D}$, normal load $W = \frac{F_t}{\cos\alpha}$; find max bending moment $M = W\frac{L}{4}$ at mid-span; compute equivalent twisting moment T_e and size shaft diameter.

1. Transmitted Torque (T) Power-to-torque relation at rated speed	$T = \frac{P \times 60}{2\pi N} = \frac{7500 \times 60}{2\pi \times 300} = \mathbf{238.7\text{ N}\cdot\text{m}}$
2. Tangential Gear Force (F_t) & Normal Load (W) Force components on gear tooth at pressure angle 20°	$F_t = \frac{2T}{D} = \frac{2 \times 238.7}{0.15} = \mathbf{3182.7\text{ N}}$$W = \frac{F_t}{\cos 20^\circ} = \frac{3182.7}{0.9397} = \mathbf{3387\text{ N}}$
3. Maximum Bending Moment (M) Mid-span bending moment for centrally-mounted gear	$M = \frac{W \cdot L}{4} = \frac{3387 \times 0.2}{4} = \mathbf{169.4\text{ N}\cdot\text{m}}$
4. Equivalent Twisting Moment (T_e) Combined bending and torsion resultant	$T_e = \sqrt{M^2 + T^2} = \sqrt{(169.4)^2 + (238.7)^2} = 292.7\text{ N}\cdot\text{m} = \mathbf{292.7 \times 10^3}$
5. Shaft Diameter (d) Torsion formula with equivalent twisting moment	$292.7 \times 10^3 = \frac{\pi}{16} \times d^3 \times 45 = 8.84d^3$$d^3 = \frac{292.7 \times 10^3}{8.84} = 33 \times 10^3$$d = 32\text{ mm}$$\Rightarrow \text{Adopt } d = \mathbf{35\text{ mm}}$

Design Output Summary

Spur Gear Shaft: P = 7.5 kW, N = 300 rpm | T = 238.7 N-m | Ft = 3182.7 N | M = 169.4 N-m | Te = 292.7 kN-mm | d = 35 mm

SECTION 7: LINE SHAFT WITH TWO SYMMETRICAL PULLEYS

System Parameters: Power $P = 100\text{ kW} = 100 \times 10^3\text{ W}$, Speed $N = 300\text{ r.p.m.}$, Shaft Span $L = 3\text{ m}$, Pulley Weight $W = 1500\text{ N}$ each, Pulley Positions 1 m from each end, Allowable Shear Stress $\tau = 60\text{N/mm}^2$ (assumed).

Design Protocol: Compute torque T ; treat shaft as simply supported beam to find $R_A = R_B = 1500\text{ N}$; find max bending moment at pulley; compute $T_e = \sqrt{M^2 + T^2}$; size shaft diameter.

1. Transmitted Torque (T) Power-speed torque conversion	$T = \frac{P \times 60}{2\pi N} = \frac{100 \times 10^3 \times 60}{2\pi \times 300} = \mathbf{3183\text{ N}\cdot\text{m}}$
2. Maximum Bending Moment (M) Simply supported beam with symmetric pulley loads	$R_A = R_B = 1500\text{ N}$$M = 1500 \times 1 = \mathbf{1500\text{ N}\cdot\text{m}}$ (at each pulley)
3. Equivalent Twisting Moment (T_e) Combined bending and torsion	$T_e = \sqrt{M^2 + T^2} = \sqrt{(1500)^2 + (3183)^2} = 3519\text{ N}\cdot\text{m} = \mathbf{3519 \times 10^3\text{ N}\cdot\text{m}}$
4. Shaft Diameter (d) Torsion formula with equivalent twisting moment	$3519 \times 10^3 = \frac{\pi}{16} \times d^3 \times 60 = 11.8d^3$$d^3 = \frac{3519 \times 10^3}{11.8} = 298 \times 10^3$$d = 66.8\text{ mm}$$\Rightarrow \text{Adopt } d = \mathbf{70\text{ mm}}$
Design Output Summary	P = 100 kW, N = 300 rpm T = 3183 N-m M = 1500 N-m at pulleys Te = 3519 kN-mm d = 70 mm

SECTION 8: OVERHUNG PULLEY SHAFT — BELT DRIVE

System Parameters: Pulley Diameter $D = 1.5\text{ m}$ ($R = 0.75\text{ m}$), Belt Tensions $T_1 = 5400\text{ N}$ (tight), $T_2 = 1800\text{ N}$ (slack), Overhang $L = 400\text{ mm}$, Allowable Shear Stress $\tau = 42\text{ MPa}$.

Design Protocol: Compute torque from belt tension difference; find total vertical load $W = T_1 + T_2$; compute overhang bending moment $M = WL$; find $T_e = \sqrt{M^2 + T^2}$; size shaft diameter.

1. Transmitted Torque (T) Net belt tension times pulley radius	$T = (T_1 - T_2)R = (5400 - 1800) \times 0.75 = \mathbf{2700 \times 10^3\text{ N}\cdot\text{mm}}$
2. Total Belt Load & Bending Moment (W, M) Sum of belt tensions creates vertical load; overhang bending	$W = T_1 + T_2 = 5400 + 1800 = \mathbf{7200\text{ N}}$ $M = W \times L = 7200 \times 400 = \mathbf{2880 \times 10^3\text{ N}\cdot\text{mm}}$
3. Equivalent Twisting Moment (T_e) Combined bending and torsion	$T_e = \sqrt{M^2 + T^2} = \sqrt{(2880 \times 10^3)^2 + (2700 \times 10^3)^2} = \mathbf{3950 \times 10^3\text{ N}\cdot\text{mm}}$
4. Shaft Diameter (d) Torsion formula for solid shaft	$3950 \times 10^3 = \frac{\pi}{16} \times d^3 \times 42 = 8.25d^3$ $d^3 = \frac{3950 \times 10^3}{8.25} = 479 \times 10^3$ $d = 78\text{ mm}$ $\Rightarrow \mathbf{\text{Adopt } d = 80\text{ mm}}$
Design Output Summary	Belt Drive Shaft: T1 = 5400 N, T2 = 1800 N, R = 0.75 m, L = 400 mm T = 2700 kN-mm M = 2880 kN-mm Te = 3950 kN-mm d = 80 mm

SECTION 9: TWO-PULLEY SHAFT — BIAXIAL BENDING AND TORSION

System Parameters: Bearing Span AB = 1 m, Pulley C: $D_C = 600$ mm, AC = 300 mm, $T_1 = 2250$ N, Pulley D: $D_D = 400$ mm, BD = 200 mm, horizontal belt; $\theta = 180^\circ$, $\mu = 0.24$, $\sigma_b = 63$ MPa, $\tau = 42$ MPa.

Design Protocol: Find T_2 from belt friction equation; compute vertical load at C and horizontal load at D; solve vertical and horizontal bending moments at C and D; find resultant M_C and M_D ; apply both failure theories; size shaft.

1. Belt Tensions on Pulley C (T_1, T_2) Belt friction formula with $\mu = 0.24$, wrap = 180°	$\log\left(\frac{T_1}{T_2}\right) = \frac{\mu\theta}{2.3} = \frac{0.24\pi}{2.3} = 0.3278 \Rightarrow \frac{T_1}{T_2} = 2.127$$T_2 = \frac{2250}{2.127} = 1058 \text{ N}$$W_C = T_1 + T_2 = 2250 + 1058 = \mathbf{3308 \text{ N (vertical)}}$
2. Torque & Belt Tensions on Pulley D (T_3, T_4) Equal torque on both pulleys; solve simultaneous equations	$T_{\text{pulley C}} = (T_1 - T_2)R_C = (2250 - 1058) \times 0.3 = 357.6 \text{ N}\cdot\text{m}$$T_3 - T_4 = \frac{357.6}{R_D} = \frac{357.6}{0.2} = 1788 \text{ N}$$\frac{T_3}{T_4} = 2.127 \Rightarrow T_3 = 3376 \text{ N}, T_4 = 1588 \text{ N}$$W_D = T_3 + T_4 = \mathbf{4964 \text{ N (horizontal)}}$
3. Vertical Bending Moments (M_{CV}, M_{DV}) Reactions and moments in vertical plane	$R_{BV} \times 1 = 3308 \times 0.3 \Rightarrow R_{BV} = 992.4 \text{ N}, \quad R_{AV} = 2315.6 \text{ N}$$M_{CV} = 2315.6 \times 0.3 = \mathbf{694.7 \text{ N}\cdot\text{m}}$$M_{DV} = 992.4 \times 0.2 = \mathbf{198.5 \text{ N}\cdot\text{m}}$
4. Horizontal Bending Moments (M_{CH}, M_{DH}) Reactions and moments in horizontal plane	$R_{BH} \times 1 = 4964 \times 0.8 \Rightarrow R_{BH} = 3971 \text{ N}, \quad R_{AH} = 993 \text{ N}$$M_{CH} = 993 \times 0.3 = \mathbf{297.9 \text{ N}\cdot\text{m}}$$M_{DH} = 3971 \times 0.2 = \mathbf{794.2 \text{ N}\cdot\text{m}}$

5. Resultant Bending Moments (M_C, M_D) Vector sum of vertical and horizontal components	$M_C = \sqrt{(694.7)^2 + (297.9)^2} = \mathbf{756\text{ N}\cdot\text{m}}$$M_D = \sqrt{(198.5)^2 + (794.2)^2} = \mathbf{819.2\text{ N}\cdot\text{m}} \rightarrow \text{Governing}$
6. Shaft Diameter — Max Shear Stress Theory (d_1) Equivalent twisting moment approach	$T_e = \sqrt{(819.2)^2 + (357.6)^2} = 894\text{ N}\cdot\text{m} = 894 \times 10^3\text{ N}\cdot\text{mm}$$894 \times 10^3 = \frac{\pi}{16} \times d^3 \times 42 = 8.25d^3$$d^3 = \frac{894 \times 10^3}{8.25} = 108 \times 10^3$$d = \mathbf{47.6\text{ mm}}$
7. Shaft Diameter — Max Normal Stress Theory (d_2) Equivalent bending moment approach	$M_e = \frac{1}{2}[819.2 + 894] = 856.6\text{ N}\cdot\text{m} = 856.6 \times 10^3\text{ N}\cdot\text{mm}$$856.6 \times 10^3 = \frac{\pi}{32} \times d^3 \times 63 = 6.2d^3$$d^3 = \frac{856.6 \times 10^3}{6.2} = 138.2 \times 10^3$$d = \mathbf{51.7\text{ mm}}$$\Rightarrow \text{Adopt } d = \mathbf{55\text{ mm}}$
Design Output Summary	Two-Pulley Shaft: M = 819.2 N-m at D, T = 357.6 N-m d1 (shear) = 47.6 mm, d2 (bending) = 51.7 mm Governing d = 55 mm

SECTION 10: SPUR GEAR AND FLYWHEEL PULLEY SHAFT

System Parameters: Bearing Span AB = 800 mm, Spur Gear *C*: $D_C = 600$ mm, $\alpha = 20^\circ$, AC = 200 mm; Pulley *D*: $D_D = 700$ mm, $W = 2000$ N, $T_1 = 3000$ N, $\frac{T_1}{T_2} = 3$, BD = 250 mm; Allowable Shear Stress $\tau = 40$ MPa.

Design Protocol: Compute torque from pulley belt tensions; find tangential and normal gear forces; resolve loads into vertical and horizontal planes; compute bending moments at *C* and *D*; find resultant moments; compute T_e and size shaft.

1. Torque & Gear Forces (T, F_{tc}, W_C) Belt-pulley torque and gear tooth force components	$T = (T_1 - T_2)R_D = 3000\left(1 - \frac{1}{3}\right) \times 350 = \mathbf{700 \times 10^3 \text{ N}\cdot\text{mm}}$ $F_{tc} = \frac{T}{R_C} = \frac{700 \times 10^3}{300} = 2333 \text{ N}$ $W_C = \frac{F_{tc}}{\cos 20^\circ} = \frac{2333}{0.9397} = 2483 \text{ N}$ $W_{CV} = 2333 \text{ N (down)}, \quad W_{CH} = 849 \text{ N}$
2. Pulley D Belt & Weight Loads (T_2, W_{DH}, W_{DV}) Belt tensions and flywheel weight at pulley <i>D</i>	$T_2 = \frac{T_1}{3} = 1000 \text{ N}$ $W_{DH} = T_1 + T_2 = 3000 + 1000 = \mathbf{4000 \text{ N}}$ $W_{DV} = W = \mathbf{2000 \text{ N (down)}}$
3. Vertical Bending Moments (M_{CV}, M_{DV}) Vertical plane: reactions and moments at <i>C</i> and <i>D</i>	$R_{BV} \times 800 = 2000(550) + 2333(200) = 1566600 \Rightarrow R_{BV} = 1958 \text{ N}, R_{AV} = 1042 \text{ N}$ $M_{CV} = 2375 \times 200 = \mathbf{475 \times 10^3 \text{ N}\cdot\text{mm}}$ $M_{DV} = 1958 \times 250 = \mathbf{489.5 \times 10^3 \text{ N}\cdot\text{mm}}$
4. Horizontal Bending Moments (M_{CH}, M_{DH}) Horizontal plane: reactions and moments at <i>C</i> and <i>D</i>	$R_{BH} \times 800 = 4000(550) + 849(200) = 2369800 \Rightarrow R_{BH} = 2963 \text{ N}, R_{AH} = 1037 \text{ N}$ $M_{CH} = 1886 \times 200 = \mathbf{377200 \text{ N}\cdot\text{mm}}$ $M_{DH} = 2963 \times 250 = \mathbf{740750 \text{ N}\cdot\text{mm}}$

5. Resultant Bending Moments & Governing Diameter (M, T_e, d) Vector sum of planes; equivalent twisting moment; shaft size	$M_D = \sqrt{(489.5 \times 10^3)^2 + (740750)^2} = 887874 \text{ N}\cdot\text{mm} \rightarrow \text{Governing}$$T_e = \sqrt{(887874)^2 + (700 \times 10^3)^2} = 1131 \times 10^3 \text{ N}\cdot\text{mm}$$1131 \times 10^3 = \frac{\pi}{16} \times d^3 \times 40 = 7.86d^3$$d^3 = \frac{1131 \times 10^3}{7.86} = 144 \times 10^3$$d = 52.4 \text{ mm}$$\Rightarrow \text{Adopt } d = 55 \text{ mm}$
Design Output Summary	Spur Gear + Pulley Shaft: Max M = 887874 N-mm at D T = 700 kN-mm Te = 1131 kN-mm d = 55 mm

SECTION 11: TWO-GEAR SHAFT — PINION AND SPUR GEAR

System Parameters: Power $P = 15$ kW, Speed $N = 200$ r.p.m., Bearing Span $AB = 750$ mm, Gear C : $T_C = 100$ teeth, $m_C = 5$ mm, $AC = 150$ mm (vertical load from below); Pinion D : $T_D = 30$ teeth, $m_D = 5$ mm, $BD = 100$ mm (horizontal load); Allowable $\tau = 54$ MPa.

Design Protocol: Compute shaft torque; find gear radii and tangential forces; solve vertical and horizontal loading planes separately; compute resultant bending moments at C and D ; find equivalent twisting moment and shaft diameter.

1. Torque & Gear Radii and Forces ($T, R_C, R_D, F_{tC}, F_{tD}$)

Power-speed torque and force distribution on each gear

$$T = \frac{P \times 60}{2\pi N} = \frac{15 \times 10^3 \times 60}{2\pi \times 200} = 716 \times 10^3 \text{ N}\cdot\text{mm}$$

$$R_C = \frac{T_C \times m_C}{2} = \frac{100 \times 5}{2} = 250 \text{ mm}, \quad R_D = \frac{30 \times 5}{2} = 75 \text{ mm}$$

$$F_{tC} = \frac{T}{R_C} = \frac{716 \times 10^3}{250} = 2870 \text{ N (down)}$$

$$F_{tD} = \frac{T}{R_D} = \frac{716 \times 10^3}{75} = 9550 \text{ N (right)}$$

2. Vertical & Horizontal Bending Moments

Separate plane analysis for bending moments at C and D

$$R_{BV} = \frac{2870 \times 150}{750} = 574 \text{ N}, \quad R_{AV} = 2296 \text{ N}$$

$$M_{CV} = 2296 \times 150 = 344400 \text{ N}\cdot\text{mm}, \quad M_{DV} = 574 \times 100 = 57400 \text{ N}\cdot\text{mm}$$

$$R_{BH} = \frac{9550 \times 650}{750} = 8277 \text{ N}, \quad R_{AH} = 1273 \text{ N}$$

$$M_{CH} = 1273 \times 150 = 190950 \text{ N}\cdot\text{mm}, \quad M_{DH} = 8277 \times 100 = 827700 \text{ N}\cdot\text{mm}$$

3. Resultant Moments & Shaft Diameter (M_C, M_D, T_e, d)

Govern from pinion location D ; equivalent twisting moment approach

$$M_D = \sqrt{(57400)^2 + (827700)^2} = \mathbf{829690\ N\cdot mm} \quad \rightarrow \text{Governing}$$

$$T_e = \sqrt{(829690)^2 + (716 \times 10^3)^2} = 1096 \times 10^3\ \text{N}\cdot\text{mm}$$

$$1096 \times 10^3 = \frac{\pi}{16} \times d^3 \times 54 = 10.6d^3$$

$$d^3 = \frac{1096 \times 10^3}{10.6} = 103.4 \times 10^3$$

$$d = 47\ \text{mm}$$

$$\Rightarrow \text{Adopt } d = 50\ \text{mm}$$

Design Output Summary

Two-Gear Shaft: P = 15 kW, N = 200 rpm | Max M = 829690 N-mm at D | T = 716 kN-mm | Te = 1096 kN-mm | d = 50 mm

SECTION 12: SHAFT DESIGN WITH SHOCK AND FATIGUE FACTORS

System Parameters: Power $P = 20$ kW, Speed $N = 200$ r.p.m., Central Load $W = 900$ N, Span $L = 2.5$ m, $\tau = 42$ MPa, $\sigma_b = 56$ MPa. Part 2: Gradually applied loads $K_m = 1.5$, $K_t = 1.0$.

Design Protocol: Find T and M (central load); apply both failure theories without shock factors; repeat with ASME shock factors for gradually applied loads; adopt larger diameter in each case.

1. Torque & Central Bending Moment (T, M) Power relation and simply-supported central load formula	$T = \frac{20 \times 10^3 \times 60}{2\pi \times 200} = \mathbf{955 \times 10^3 \text{ N}\cdot\text{mm}}$$M = \frac{WL}{4} = \frac{900 \times 2500}{4} = \mathbf{562.5 \times 10^3 \text{ N}\cdot\text{mm}}$
2. Diameter — Case 1: Steady Loads (Max Shear Stress) Equivalent twisting moment without shock factors	$T_e = \sqrt{(562.5 \times 10^3)^2 + (955 \times 10^3)^2} = \mathbf{1108 \times 10^3 \text{ N}\cdot\text{mm}}$$1108 \times 10^3 = \frac{\pi}{16} \times d^3 \times 42 = 8.25d^3$$d^3 = 134.3 \times 10^3 \Rightarrow d = \mathbf{51.2 \text{ mm}}$
3. Diameter — Case 1: Steady Loads (Max Normal Stress) Equivalent bending moment without shock factors	$M_e = \frac{1}{2}[562.5 + 1108] \times 10^3 = \mathbf{835.25 \times 10^3 \text{ N}\cdot\text{mm}}$$835.25 \times 10^3 = \frac{\pi}{32} \times d^3 \times 56 = 5.5d^3$$d^3 = 152 \times 10^3 \Rightarrow d = 53.4 \text{ mm}$$\Rightarrow \mathbf{\text{Adopt } d = 55 \text{ mm}}$
4. Diameter — Case 2: Gradually Applied ($K_m = 1.5$, $K_t = 1.0$), Max Shear Stress ASME shock factors applied to bending and torsion	$T_e = \sqrt{(1.5 \times 562.5 \times 10^3)^2 + (1.0 \times 955 \times 10^3)^2} = \mathbf{1274 \times 10^3 \text{ N}\cdot\text{mm}}$$1274 \times 10^3 = 8.25d^3 \Rightarrow d^3 = 154.6 \times 10^3 \Rightarrow d = \mathbf{53.6 \text{ mm}}$

5. Diameter — Case 2: Gradually Applied, Max Normal Stress

Equivalent bending moment with shock factors

$$M_e = \frac{1}{2} [1.5 \times 562.5 \times 10^3 + 1274 \times 10^3] = \mathbf{1059 \times 10^3 \text{ N}\cdot\text{mm}}$$
$$1059 \times 10^3 = 5.5d^3 \Rightarrow d^3 = 192.5 \times 10^3 \Rightarrow d = 57.7 \text{ mm}$$

$\Rightarrow$ **Adopt $d = 60 \text{ mm}$**

Design Output Summary

Steady Loads: d = 55 mm | Gradually Applied (Km=1.5, Kt=1.0): d = 60 mm

SECTION 13: LATHE SHAFT — BELT DRIVE WITH SHOCK FACTORS

System Parameters: Pulley Weight $W = 200$ N, Overhang $L = 300$ mm, Pulley Diameter $D = 200$ mm ($R = 100$ mm), Power $P = 1$ kW, Speed $N = 120$ r.p.m., $\theta = 180^\circ$, $\mu = 0.3$, $K_m = 1.5$, $K_t = 2.0$, $\tau = 35$ MPa.

Design Protocol: Compute torque; solve belt tensions T_1 and T_2 from friction and torque equations; find total vertical load on pulley; compute bending moment; apply shock factors to T_e ; size shaft diameter.

1. Transmitted Torque (T) Power-to-torque conversion	$T = \frac{1000 \times 60}{2\pi \times 120} = 79.6 \text{ N}\cdot\text{m} = \mathbf{79.6 \times 10^3 \text{ N}\cdot\text{mm}}$
2. Belt Tensions (T_1, T_2) Belt friction and torque equations solved simultaneously	$T_1 - T_2 = \frac{79.6 \times 10^3}{100} = 796 \text{ N}$$\log\left(\frac{T_1}{T_2}\right) = \frac{\mu\theta}{2.3} = \frac{0.3\pi}{2.3} = 0.4098 \Rightarrow \frac{T_1}{T_2} = 2.57$$T_1 = \mathbf{1303 \text{ N}}, \quad T_2 = \mathbf{507 \text{ N}}$
3. Total Load & Bending Moment (W_T, M) Combined belt tension and pulley weight; overhang bending	$W_T = T_1 + T_2 + W = 1303 + 507 + 200 = \mathbf{2010 \text{ N}}$$M = W_T \times L = 2010 \times 300 = \mathbf{603 \times 10^3 \text{ N}\cdot\text{mm}}$
4. Shaft Diameter (d) Equivalent twisting moment with shock factors	$T_e = \sqrt{(K_m M)^2 + (K_t T)^2} = \sqrt{(1.5 \times 603 \times 10^3)^2 + (2 \times 79.6 \times 10^3)^2}$$918 \times 10^3 = \frac{\pi}{16} \times d^3 \times 35 = 6.87d^3$$d^3 = \frac{918 \times 10^3}{6.87} = 133.6 \times 10^3$$d = 51.1 \text{ mm}$$\Rightarrow \text{Adopt } d = \mathbf{55 \text{ mm}}$

Design Output Summary

Lathe Shaft: T = 79.6 kN-mm | T1 = 1303 N, T2 = 507 N | M = 603 kN-mm | Te = 918 kN-mm (Km=1.5, Kt=2.0) | d = 55 mm

SECTION 14: PULLEY AND GEAR SHAFT — ASME SHOCK FACTORS

System Parameters: Power $P = 20\text{ kW}$, Speed $N = 150\text{ r.p.m.}$, $\frac{T_1}{T_2} = 2.5$, Gear Weight $W_B = 900\text{ N}$, Pulley Weight $W_A = 2700\text{ N}$, $\tau = 63\text{ MPa}$, $K_m = 2$, $K_t = 1.5$, Gear $D_B = 750\text{ mm}$ ($R_B = 375\text{ mm}$), Pulley $D_A = 1250\text{ mm}$ ($R_A = 625\text{ mm}$), Bearing Span $CD = 1000\text{ mm}$, $AD = 1250\text{ mm}$, $BC = 350\text{ mm}$.

Design Protocol: Compute torque; find belt tensions T_1, T_2 ; compute total vertical loads at A and B ; take moments to find reactions R_C, R_D ; find bending moments at C and D ; apply shock factors to T_e and size shaft.

1. Torque & Belt Tensions (T, T_1, T_2) Power relation and belt friction with $\frac{T_1}{T_2} = 2.5$	$T = \frac{20 \times 10^3 \times 60}{2\pi \times 150} = 1273 \times 10^3\text{ N}\cdot\text{mm}$$T_1 - T_2 = \frac{1273 \times 10^3}{625} = 2037\text{ N}, \quad T_1 = 2.5T_2$$T_2 = 1358\text{ N}, \quad T_1 = 3395\text{ N}$
2. Loads at A and B, Reactions & Bending Moments Vertical loading: downward at A, upward net at B; bending moments	Load at A = $T_1 + T_2 + W_A = 3395 + 1358 + 2700 = \mathbf{7453\text{ N}}$ (down) Net at B = $F_t - W_B = \frac{1273 \times 10^3}{375} - 900 = 3395 - 900 = \mathbf{2495\text{ N}}$ (up) $R_C \times 1000 = 7453 \times 1250 + 2495 \times 350 = 10.2 \times 10^6 \Rightarrow R_C = 10200\text{ N}$ $M_C = 7453 \times 250 = \mathbf{1863 \times 10^3\text{ N}\cdot\text{mm}}$ $\rightarrow$ Governing
3. Shaft Diameter (d) Equivalent twisting moment with ASME shock factors $K_m = 2, K_t = 1.5$	$T_e = \sqrt{(2 \times 1863 \times 10^3)^2 + (1.5 \times 1273 \times 10^3)^2} = \mathbf{4187 \times 10^3\text{ N}\cdot\text{mm}}$$4187 \times 10^3 = \frac{\pi}{16} \times d^3 \times 63 = 12.37d^3$$d^3 = \frac{4187 \times 10^3}{12.37} = 338 \times 10^3$$d = 69.6\text{ mm}$$\Rightarrow \text{Adopt } d = \mathbf{70\text{ mm}}$

Design Output Summary

Pulley+Gear Shaft: T = 1273 kN-mm | Max M = 1863 kN-mm at C | Te = 4187 kN-mm
(Km=2, Kt=1.5) | d = 70 mm

SECTION 15: NICKEL STEEL SHAFT — DUAL VERTICAL GEAR LOADING

System Parameters: Bearing Span AB = 2400 mm, Gear *C*: $D_C = 600$ mm, AC = 250 mm, $W_C = 950$ N; Gear *D*: $D_D = 200$ mm, BD = 400 mm, $W_D = 350$ N; $P = 20$ kW, $N = 120$ r.p.m., $\sigma_t = 100$ MPa, $\tau = 56$ MPa, $K_m = 1.5$, $K_t = 1.2$.

Design Protocol: Compute torque; find tangential forces at *C* and *D* plus gear weights; take moments for reactions; find M_C and M_D ; apply both failure theories with shock factors; govern from larger diameter.

1. Torque & Tangential Forces (T, F_{tC}, F_{tD}) Power torque and gear tooth loads at each gear location	$T = \frac{20 \times 10^3 \times 60}{2\pi \times 120} = 1590 \times 10^3 \text{ N}\cdot\text{mm}$$F_{tC} = \frac{1590 \times 10^3}{300} = 5300 \text{ N} \Rightarrow \text{Total at } C = 5300 + 950 = \mathbf{6250 \text{ N}}$$F_{tD} = \frac{1590 \times 10^3}{100} = 15900 \text{ N} \Rightarrow \text{Total at } D = 15900 + 350 = \mathbf{16250 \text{ N}}$
2. Reactions & Bending Moments (R_A, R_B, M_C, M_D) Statics of simply supported beam with two downward loads	$R_B \times 2400 = 16250 \times 2000 + 6250 \times 250 = 34062.5 \times 10^3$$R_B = 14190 \text{ N}, \quad R_A = 22500 - 14190 = 8310 \text{ N}$$M_C = 8310 \times 250 = 2077.5 \times 10^3 \text{ N}\cdot\text{mm}$$M_D = 14190 \times 400 = \mathbf{5676 \times 10^3 \text{ N}\cdot\text{mm}} \rightarrow \text{Governing}$
3. Diameter — Max Shear Stress Theory (d_1) Equivalent twisting moment with shock factors	$T_e = \sqrt{(1.5 \times 5676 \times 10^3)^2 + (1.2 \times 1590 \times 10^3)^2} = \mathbf{8725 \times 10^3}$$8725 \times 10^3 = \frac{\pi}{16} \times d^3 \times 56 = 11d^3$$d^3 = \frac{8725 \times 10^3}{11} = 793 \times 10^3 \Rightarrow d = \mathbf{92.5 \text{ mm}}$

4. Diameter — Max Normal Stress Theory (d_2)

Equivalent bending moment with shock factors

$$M_e = \frac{1}{2} [1.5 \times 5676 \times 10^3 + 8725 \times 10^3] = 8620 \times 10^3 \text{ N}\cdot\text{mm}$$
$$8620 \times 10^3 = \frac{\pi}{32} \times d^3 \times 100 = 9.82d^3$$
$$d^3 = \frac{8620 \times 10^3}{9.82} = 878 \times 10^3 \Rightarrow d = 95.7 \text{ mm}$$

$\Rightarrow$ Adopt $d = 100 \text{ mm}$

Design Output Summary

Nickel Steel Shaft: Max M = 5676 kN-mm at D | T = 1590 kN-mm | d1 = 92.5 mm, d2 = 95.7 mm | Governing d = 100 mm

SECTION 16: HOISTING DRUM SHAFT — MOTOR, TORQUE AND DIAMETER

System Parameters: Drum Diameter $D = 0.5\text{ m}$ ($R = 0.25\text{ m}$), Reduction Ratio = 12 : 1, Hoist Load $W = 8000\text{ N}$, Hoisting Speed $v = 50\text{ m/min}$, Drive Efficiency $\eta = 0.80$, Drive Gear $D_1 = 450\text{ mm}$ ($R_1 = 225\text{ mm}$), Overhang = 150 mm, $\sigma_t = 115\text{ MPa}$, $\tau = 50\text{ MPa}$, Pressure Angle = 20° , $K_m = 2$, $K_t = 1.5$.

Design Protocol: Compute drum power and motor power; find drum torque $T = WR$; find motor speed from reduction ratio; compute bending load from overhang gear; apply both failure theories with shock factors; adopt larger diameter.

1. Motor Power & Drum Torque (P, T) Energy rate and torque at drum shaft	$P_{\text{drum}} = \frac{W \times v}{60} = \frac{8000 \times 50}{60} = 6670\text{ W}$$P_{\text{motor}} = \frac{6670}{0.8} = \mathbf{8.33\text{ kW}}$$T = W \times R = 8000 \times 0.25 = \mathbf{2000\text{ N}\cdot\text{m}}$
2. Motor Speed (N_{motor}) Speed from drum radius and reduction ratio	$\omega_{\text{drum}} = \frac{v}{R} = \frac{50}{0.25} = 200\text{ rad/min}$$\omega_{\text{motor}} = 200 \times 12 = 2400\text{ rad/min}$$N_{\text{motor}} = \frac{2400}{2\pi} = \mathbf{382\text{ r.p.m.}}$
3. Bending Moment from Overhang Gear (F_t, W_b, M) Normal tooth load and overhang bending moment at bearing	$F_t = \frac{T}{R_1} = \frac{2000}{0.225} = 8900\text{ N}$$W_b = \frac{F_t}{\cos 20^\circ} = \frac{8900}{0.9397} = 9470\text{ N}$$M = 9470 \times 0.15 = \mathbf{1420\text{ N}\cdot\text{m}}$

4. Diameter — Max Shear Stress Theory (d_1)

Equivalent twisting moment approach

$$T_e = \sqrt{(2 \times 1420)^2 + (1.5 \times 2000)^2} = 4130 \text{ N}\cdot\text{m} = \mathbf{4130 \times 10^3 \text{ N}\cdot\text{mm}}$$
$$4130 \times 10^3 = \frac{\pi}{16} \times d^3 \times 50 = 9.82d^3$$
$$d^3 = \frac{4130 \times 10^3}{9.82} = 420.6 \times 10^3 \Rightarrow d = \mathbf{75 \text{ mm}}$$

5. Diameter — Max Normal Stress Theory (d_2)

Equivalent bending moment approach

$$M_e = \frac{1}{2}[2 \times 1420 + 4130] = 3485 \text{ N}\cdot\text{m} = \mathbf{3485 \times 10^3 \text{ N}\cdot\text{mm}}$$
$$3485 \times 10^3 = \frac{\pi}{32} \times d^3 \times 115 = 11.3d^3$$
$$d^3 = \frac{3485 \times 10^3}{11.3} = 308.4 \times 10^3 \Rightarrow d = \mathbf{67.5 \text{ mm}}$$

$\Rightarrow$ **Governing** $d = \mathbf{75 \text{ mm}}$ (from Max Shear Stress Theory)

Design Output Summary

Hoisting Drum: Motor P = 8.33 kW, N = 382 rpm | Drum T = 2000 N-m | M = 1420 N-m | Te = 4130 kN-mm | d = 75 mm

SECTION 17: COMPLEX SHAFT — GEAR AND TWO PULLEYS, BIAXIAL LOADING

System Parameters: Bearing Span PQ = 1.8 m, Gear *D*: $D_D = 300$ mm, $\alpha = 20^\circ$, QD = 150 mm, $P_D = 30$ kW input; Pulley *B*: $D_B = 750$ mm, $P_B = 11.25$ kW, vertical belt, $\frac{T_{B1}}{T_{B2}} = 2$; Pulley *C*: $D_C = 600$ mm, $P_C = 18.75$ kW, belt at 60° , $\frac{T_{C1}}{T_{C2}} = 2$; Speed $N = 250$ r.p.m., $K_m = 2$, $K_t = 1.5$, $\tau = 42$ MPa, $\sigma_t = 84$ MPa.

Design Protocol: Compute individual torques for gear and pulleys; find belt tensions and resolve forces vertically/horizontally at each location; compute bending moments in both planes; find resultant moments; apply shock factors; govern by larger diameter from two failure theories.

1. Gear D Forces ($T_D, F_{tD}, W_{DV}, W_{DH}$) Gear torque and normal load components at pressure angle 20°	$T_D = \frac{30 \times 10^3 \times 60}{2\pi \times 250} = 1146 \text{ N}\cdot\text{m}$$F_{tD} = \frac{1146}{0.15} = 7640 \text{ N}, \quad W_D = \frac{7640}{\cos 20^\circ} = 8130 \text{ N}$$W_{DV} = 8130 \cos 20^\circ = \mathbf{7640 \text{ N}}, \quad W_{DH} = 8130 \sin 20^\circ = \mathbf{2780 \text{ N}}$
2. Pulley C Belt Forces ($T_C, T_{C1}, T_{C2}, W_{CV}, W_{CH}$) Belt at 60° to horizontal; resolve total belt load	$T_C = \frac{18.75 \times 10^3 \times 60}{2\pi \times 250} = 716 \text{ N}\cdot\text{m}$$T_{C1} - T_{C2} = \frac{716}{0.3} = 2387 \text{ N}, \quad \frac{T_{C1}}{T_{C2}} = 2 \Rightarrow T_{C2} = 2387, T_{C1} = 4774 \text{ N}$$W_C = 4774 + 2387 = 7161 \text{ N}$$W_{CV} = 7161 \sin 60^\circ = \mathbf{6200 \text{ N}}, \quad W_{CH} = 7161 \cos 60^\circ = \mathbf{3580 \text{ N}}$
3. Pulley B Belt Forces ($T_B, T_{B1}, T_{B2}, W_{BV}$) Vertical belt; total load acts downward	$T_B = \frac{11.25 \times 10^3 \times 60}{2\pi \times 250} = 430 \text{ N}\cdot\text{m}$$T_{B1} - T_{B2} = \frac{430}{0.375} = 1147 \text{ N}, \quad T_{B2} = 1147 \text{ N}, T_{B1} = 2294 \text{ N}$$W_{BV} = T_{B1} + T_{B2} = \mathbf{3441 \text{ N (down)}}, \quad W_{BH} = 0$

4. Vertical & Horizontal Bending Moments

Reactions and moments in each plane at B, C, D

$$R_{QV} \times 1.8 = 7640(1.65) + 6200(1.35) + 3441(0.6) = 23041 \Rightarrow R_{QV} = 12799 \text{ N}$$

$$M_{BV} = 4481(0.6) = 2690 \text{ N}\cdot\text{m}, \quad M_{CV} = 4481(1.35) - 3441(0.75) = 3470 \text{ N}\cdot\text{m}$$

$$R_{QH} \times 1.8 = 2780(1.65) + 3580(1.35) = 9420 \Rightarrow R_{QH} = 5233 \text{ N}$$

$$M_{BH} = 1127(0.6) = 676 \text{ N}\cdot\text{m}, \quad M_{CH} = 1127(1.35) = 1521 \text{ N}\cdot\text{m}$$

5. Resultant Moments & Governing Diameter (M_C, T_e, d)

Resultant at each point; max shear stress theory with shock factors

$$M_C = \sqrt{(3470)^2 + (1521)^2} = \mathbf{3790 \text{ N}\cdot\text{m}} \quad \rightarrow \text{Governing}$$

$$T_e = \sqrt{(2 \times 3790)^2 + (1.5 \times 1146)^2} = 7772 \text{ N}\cdot\text{m} = \mathbf{7772 \times 10^3 \text{ N}\cdot\text{mm}}$$

$$7772 \times 10^3 = \frac{\pi}{16} \times d^3 \times 42 = 8.25d^3$$

$$d^3 = \frac{7772 \times 10^3}{8.25} = 942 \times 10^3 \Rightarrow d = 98 \text{ mm}$$

$$M_e = \frac{1}{2}[2 \times 3790 + 7772] = 7676 \text{ N}\cdot\text{m}$$

$$7676 \times 10^3 = 8.25d^3 \Rightarrow d^3 = 930 \times 10^3 \Rightarrow d = 97.6 \text{ mm}$$

$$\Rightarrow \text{Adopt } d = 100 \text{ mm}$$

Design Output Summary

Complex Shaft: Governing M = 3790 N-m at C | T = 1146 N-m | Te = 7772 kN-mm |
Me = 7676 kN-mm | d = 100 mm

SECTION 18: HOLLOW SHAFT — COMBINED TORQUE, BENDING, AND AXIAL LOAD

System Parameters: Torque $T = 1.5 \text{ kN}\cdot\text{m}$, Bending Moment $M = 3 \text{ kN}\cdot\text{m}$, Axial Load $F = 10 \text{ kN}$ (tensile), Hollow Ratio $k = \frac{d_i}{d_o} = 0.5$, Outside Diameter $d_o = 80 \text{ mm}$, Load Applied Gradually ($K_m = 1.5$, $K_t = 1.0$, $\alpha = 1$). Find induced shear stress τ .

Design Protocol: Substitute given values into hollow shaft equivalent twisting moment formula including axial load term; solve for induced shear stress from torsion formula for hollow shaft.

1. Equivalent Twisting Moment — Hollow Shaft (T_e) Combined bending, axial, and torsion for hollow section	$T_e = \sqrt{\left[K_m M + \frac{\alpha F d_o (1 + k^2)}{8}\right]^2 + (K_t T)^2}$$= \sqrt{\left[1.5(3 \times 10^3) + \frac{1 \times 10 \times 10^3 \times 0.08(1 + 0.25)}{8}\right]^2 + (1 \times 1.5 \times 10^3)^2}$$= \sqrt{(4500 + 125)^2 + (1500)^2}$$= 4862 \text{ N}\cdot\text{m} = 4862 \times 10^3 \text{ N}\cdot\text{mm}$
2. Induced Shear Stress (τ) Hollow shaft torsion formula with $k = 0.5$, $d_o = 80 \text{ mm}$	$T_e = \frac{\pi}{16} \times \tau \times (d_o)^3 (1 - k^4)$$4862 \times 10^3 = \frac{\pi}{16} \times \tau \times (80)^3 [1 - (0.5)^4] = 94260\tau$$\tau = \frac{4862 \times 10^3}{94260}$$= 51.6\text{N/mm}^2 = 51.6 \text{ MPa}$
Design Output Summary	Hollow Shaft: T = 1.5 kN-m, M = 3 kN-m, F = 10 kN axial k = 0.5, do = 80 mm Te = 4862 kN-mm Induced tau = 51.6 MPa

SECTION 19: MARINE PROPELLER HOLLOW SHAFT — SHEAR STRESS AND TWIST

System Parameters: Outside Diameter $d_o = 0.5$ m, Inside Diameter $d_i = 0.3$ m, $k = \frac{d_i}{d_o} = 0.6$, Power $P = 5600$ kW, Speed $N = 150$ r.p.m., Bearing Span $L = 6$ m, Axial Thrust $F = 500$ kN, Shaft Weight $W = 70$ kN, $G = 84$ GPa.

Design Protocol: Compute torque and max bending moment; calculate slenderness ratio and column factor α ; use hollow shaft T_e formula; find induced shear stress; compute polar moment of inertia J and angular twist θ .

1. Torque & Maximum Bending Moment (T, M)

Power-speed torque and uniform weight bending moment

$$T = \frac{5600 \times 10^3 \times 60}{2\pi \times 150} = \mathbf{356460 \text{ N}\cdot\text{m}}$$
$$M = \frac{WL}{8} = \frac{70 \times 10^3 \times 6}{8} = \mathbf{52500 \text{ N}\cdot\text{m}}$$

2. Slenderness Ratio & Column Factor ($K, \frac{L}{K}, \alpha$)

Radius of gyration and column factor for axial load correction

$$K = \frac{1}{4} \sqrt{(d_o)^2 + (d_i)^2} = \frac{1}{4} \sqrt{(0.5)^2 + (0.3)^2} = 0.1458 \text{ m}$$
$$\frac{L}{K} = \frac{6}{0.1458} = 41.15$$
$$\alpha = \frac{1}{1 - 0.0044(\frac{L}{K})} = \frac{1}{1 - 0.0044 \times 41.15} = \mathbf{1.22}$$

3. Equivalent Twisting Moment & Shear Stress (T_e, τ)

Hollow shaft with axial load, $K_m = 1.5$, $K_t = 1.0$

$$T_e = \sqrt{\left[1.5(52500) + \frac{1.22(500 \times 10^3)(0.5)(1 + 0.6^2)}{8}\right]^2 + (356460)^2}$$
$$= \sqrt{(78750 + 51850)^2 + (356460)^2} = \mathbf{380 \times 10^3 \text{ N}\cdot\text{m}}$$
$$380 \times 10^3 = \frac{\pi}{16} \times \tau \times (0.5)^3 [1 - (0.6)^4] = 0.02\tau$$
$$\tau = \frac{380 \times 10^3}{0.02} = \mathbf{19 \text{ MPa}}$$

4. Angular Twist Between Bearings (J, θ)

Polar moment of inertia and torsion angle

$$J = \frac{\pi}{32} \left[(d_o)^4 - (d_i)^4 \right] = \frac{\pi}{32} [(0.5)^4 - (0.3)^4] = 0.00534 \text{m}^4$$
$$\theta = \frac{T \cdot L}{G \cdot J} = \frac{356460 \times 6}{84 \times 10^9 \times 0.00534} = 0.0048 \text{ rad}$$
$$= 0.0048 \times \frac{180}{\pi} = \mathbf{0.275^\circ}$$

Design Output Summary

Marine Propeller Shaft: do = 0.5 m, di = 0.3 m | T = 356460 N-m | Te = 380 kN-m |
Induced tau = 19 MPa | Twist = 0.275 deg

SECTION 20: HOLLOW SHAFT — CYCLIC LOADS WITH AXIAL COMPRESSION

System Parameters: $M = 1000 \text{ N}\cdot\text{m}$, $T = 500 \text{ N}\cdot\text{m}$, Axial Compressive Load $F = 15 \text{ kN}$, Span $L = 1500 \text{ mm}$, $\tau = 40 \text{ MPa}$, $k = \frac{d_i}{d_o} = 0.8$, Shock Factors $K_t = 1.5$, $K_m = 1.6$; shock applied load.

Design Protocol: Express radius of gyration $K = 0.32d_o$; write column factor α in terms of d_o ; set up T_e equation; match against torsion capacity $4.65(d_o)^3$; solve by trial and iteration.

1. Radius of Gyration & Column Factor (K, α) Hollow shaft K in terms of d_o; column factor as function of d_o	$K = \frac{\sqrt{(d_o)^2 + (d_i)^2}}{4} = \frac{d_o \sqrt{1 + 0.64}}{4} = 0.32d_o$$\alpha = \frac{1}{1 - 0.0044\left(\frac{1500}{0.32}d_o\right)} = \frac{1}{1 - \frac{20.6}{d_o}}$
2. Torsional Capacity of Hollow Shaft ($T_e = 4.65(d_o)^3$) RHS torsion equation, $k = 0.8$	$T_e = \frac{\pi}{16} \times 40 \times (d_o)^3 (1 - 0.8^4) = 4.65(d_o)^3$
3. Trial Solution — Hollow Shaft Diameters (d_o, d_i) Hit-and-trial: neglect axial gives $d_o > 72.4 \text{ mm}$; trial at $d_o = 76.32 \text{ mm}$	$\text{Min bound (no axial): } = T_e = \sqrt{(1.6 \times 10^6)^2 + (0.75 \times 10^6)^2} = 1767 \times$$4.65(d_o)^3 = 1767 \times 10^3 \Rightarrow d_o = 72.4 \text{ mm (min)}$$\text{Trial: } d_o = 76.32 \text{ mm} \Rightarrow d_o = 80 \text{ mm}$$d_i = 0.8 \times 80 = 64 \text{ mm}$
Design Output Summary	Hollow Shaft: M = 1000 N-m, T = 500 N-m, F = 15 kN compressive k = 0.8 Km = 1.6, Kt = 1.5 do = 80 mm, di = 64 mm

SECTION 21: SPINDLE DIAMETER — ANGULAR DEFLECTION LIMIT

System Parameters: Power $P = 4\text{ kW} = 4000\text{ W}$, Speed $N = 800\text{ r.p.m.}$, Max Angular Deflection $\theta = 0.25^\circ\text{ per metre} = 0.0044\text{ rad per }1000\text{ mm}$, Modulus of Rigidity $G = 84\text{ GPa} = 84 \times 10^3\text{N/mm}^2$.

Design Protocol: Compute torque; equate torsion rigidity equation $\frac{T}{J} = G \frac{\theta}{L}$ to find polar moment of inertia J ; solve for spindle diameter; then find actual induced shear stress.

1. Transmitted Torque (T) Power-speed torque relation	$T = \frac{P \times 60}{2\pi N} = \frac{4000 \times 60}{2\pi \times 800} = 47.74\text{ N}\cdot\text{m} = \mathbf{47740\text{ N}\cdot\text{mm}}$
2. Spindle Diameter from Stiffness Condition (d) Rigidity equation: $\frac{T}{J} = G \frac{\theta}{L}$, solve for d	$\frac{\pi}{32}d^4 = \frac{T \cdot L}{G \cdot \theta} = \frac{47740 \times 1000}{84 \times 10^3 \times 0.0044} = 129167$ $d^4 = \frac{129167 \times 32}{\pi} = 1.3 \times 10^6$ $d = 33.87\text{ mm}$ $\Rightarrow \mathbf{\text{Adopt } d = 35\text{ mm}}$
3. Induced Shear Stress (τ) Torsion formula with adopted diameter $d = 35\text{ mm}$	$T = \frac{\pi}{16} \times d^3 \times \tau \Rightarrow 47740 = \frac{\pi}{16} \times (35)^3 \times \tau = 8420\tau$ $\tau = \frac{47740}{8420} = \mathbf{5.67\text{N/mm}^2} = \mathbf{5.67\text{ MPa}}$
Design Output Summary	Steel Spindle: P = 4 kW, N = 800 rpm, theta_limit = 0.25 deg/m T = 47740 N-mm d = 35 mm Induced tau = 5.67 MPa

SECTION 22: HOLLOW VS SOLID SHAFT — WEIGHT, STRENGTH, STIFFNESS RATIO

System Parameters: Both shafts have same external diameter $d_o = d$, same material and length. Hollow shaft inside diameter $d_i = \frac{d_o}{2}$, so $k = \frac{d_i}{d_o} = 0.5$. Compare weight ($\frac{W_H}{W_S}$), torsional strength ($\frac{T_H}{T_S}$), and torsional stiffness ($\frac{S_H}{S_S}$).

Design Protocol: Express cross-sectional area ratio for weight; polar section modulus ratio for strength; polar moment of inertia ratio for stiffness; evaluate each ratio using $k = 0.5$.

1. Weight Comparison ($\frac{W_H}{W_S}$) Cross-sectional area ratio for equal length and density	$\frac{W_H}{W_S} = \frac{(d_o)^2 - (d_i)^2}{d^2} = 1 - k^2 = 1 - (0.5)^2 = \mathbf{0.75}$
2. Strength Comparison ($\frac{T_H}{T_S}$) Ratio of polar section moduli (torsional capacity)	$\frac{T_H}{T_S} = \frac{(d_o)^3(1 - k^4)}{d^3} = 1 - k^4 = 1 - (0.5)^4 = \mathbf{0.9375}$
3. Stiffness Comparison ($\frac{S_H}{S_S}$) Ratio of polar moments of inertia $\frac{J_H}{J_S}$	$\frac{S_H}{S_S} = \frac{J_H}{J_S} = \frac{(d_o)^4 - (d_i)^4}{d^4} = 1 - k^4 = 1 - (0.5)^4 = \mathbf{0.9375}$
Summary of Hollow vs Solid Shaft ($k = 0.5$)	Weight ratio WH/WS = 0.75 (25% lighter) Strength ratio TH/TS = 0.9375 (6.25% weaker) Stiffness ratio SH/SS = 0.9375 (6.25% less stiff)